

TriChannel - complete results compendium

Every table below is rendered at generation time from the JSON files shipped in results/; the figures use the same plotting code as figs/make_figs.py and figs/make_figs2.py. Percentages are exact to the precision shown; intervals are 95% paired-bootstrap.

Headline numbers

action-shaped gain (AUC, fused vs best single)	0.832 -> 0.874
action-shaped gain (TPR@7% FPR)	35.4% -> 50.8%
off-regime repair, InjecAgent (max -> LR)	0.944 -> 0.978
cross-bench transfer, toolcall -> InjecAgent	0.9766 (best single 0.9718)
adaptive fused AUC (GLM-4.6)	0.795 [0.766, 0.825]
adaptive margin over text (paired CI)	[0.011, 0.026]
signature-evading draws (circularity)	307/440; fused 26.4% vs text 25.1%
contamination pre -> post (GLM-4.6 text AUC)	0.518 -> 0.915
enforcement layer (irreversible prevented)	100% (78/90 attacks)

Regime baseline: no single channel covers both shapes

detector	InjecAgent TPR	tool-call TPR	AgentDojo TPR
GLM-4.6 (prompted)	79.1%	23.8%	2070.0%
GLM-5.2	95.2%	34.6%	4360.0%
GPT-4o-mini	97.9%	27.1%	8430.0%
Claude Haiku 4.5	89.3%	34.6%	3430.0%
Llama Guard 4 (trained)	46.0%	10.8%	6960.0%
full CaMeL (provenance)	4.3%	42.5%	20.7%
signature layer (pattern)	0.0%	70.0%	9.6%

Sources: run_lg4_comparison.json, run_or_extra_comparison.json, run_camel_full_eval.json, run_toolcall_bench.json, run_agentdojo.json. Signature layer on AgentDojo = det-pipeline TPR; CaMeL AgentDojo has no benign control.

Tool-call bench, matched-FPR operating points (TPR %)

detector	AUC	@1%	@5%	@7%	@10%
GLM-4.6	0.642	28.3	28.3	28.3	28.3
GLM-5.2	0.764	35.8	44.2	49.6	49.6
GPT-4o-mini	0.664	28.7	35.0	35.0	35.0
Claude Haiku 4.5	0.832	35.4	35.4	35.4	35.4
det (non-conf)	0.656	31.2	31.2	31.2	31.2

Source: run_toolcall_matched_fpr.json.

AgentDojo matched-FPR comparators

comparator	AUC	TPR@0.5	FPR@0.5	TPR@67.5%FPR
glm-5.2	0.845	4360.0%	750.0%	8860.0%
gpt-4o-mini	0.837	8430.0%	1330.0%	8500.0%
claude-haiku-4-5	0.924	3430.0%	0.0%	9390.0%

System: TPR 92.5% at FPR 67.5%. Source: run_agentdojo_matched_fpr.json.

Channel-subset ablation grid (max-fusion)

subset	InjA AUC	InjA TPR	TC AUC	TC TPR	AD AUC	AD TPR
text	0.972	94.1%	0.832	35.4%	0.923	52.5%
provenance	0.503	4.3%	0.713	42.5%	-	-
pattern	0.500	0.0%	0.656	31.2%	0.152	0.0%
text+provenance	0.944	95.7%	0.845	42.5%	-	-
text+pattern	0.972	94.1%	0.861	44.6%	0.237	0.0%
provenance+pattern	0.503	4.3%	0.754	50.8%	-	-
text+provenance+pattern	0.944	95.7%	0.874	50.8%	-	-

TPR at 7% FPR. n = malicious/benign per bench: InjecAgent 187/323, tool-call 240/240, AgentDojo 280/120.
Source: run_ensemble_ablation.json.

Tighter operating points and score granularity

budget	threshold	actual FPR	blocking tie	benign at tie
1pct	0.18	0.42%	0.15	29
7pct	0.18	0.42%	0.15	29

Text guard emits 18 distinct scores over 480 trajectories; the benign tie plateau at 0.15 blocks both budgets, so TPR@1% = TPR@7% for every subset. Source: run_tpr_at_1pct.json.

Learned linear head (5-fold CV)

bench	max-fusion AUC	LR CV AUC	coef (text, prov, pattern)
toolcall	0.874	0.8751 +/- 0.0220	1.76, 1.50, 1.50
injecagent	0.944	0.9776 +/- 0.0196	4.03, 0.31, 0.00

Fusion-rule comparison

rule	InjecAgent AUC [CI]	tool-call AUC [CI]
max	0.9441 [0.916, 0.970]	0.8737 [0.845, 0.901]
mean	0.9446 [0.917, 0.970]	0.8737 [0.845, 0.901]
noisy-or	0.9441 [0.916, 0.970]	0.8737 [0.845, 0.901]
product	0.5000 [0.500, 0.500]	0.6146 [0.589, 0.641]
lr_oof	0.9746 [0.954, 0.991]	0.8695 [0.838, 0.898]

Source: run_fusion_rule_ablation.json; lr_oof = pooled out-of-fold decision function.

Cross-benchmark transfer

train -> evaluate	LR AUC	max-fusion	best single
toolcall -> injecagent	0.9766	0.9441	0.9718
toolcall -> agentdojo	n/a	n/a	n/a
injecagent -> toolcall	0.8447	0.8737	0.8316
injecagent -> agentdojo	n/a	n/a	n/a

Composition ceiling and benchmark contamination

Instruction-shape filter composed with each classifier (InjecAgent, real benign; max-fusion)

classifier	alone	composed	diff	95% CI
GLM-4.6	0.915	0.769	-0.146	[-0.186, -0.104]
GLM-5.2	0.997	0.757	-0.241	[-0.281, -0.201]
Claude Haiku 4.5	0.972	0.766	-0.205	[-0.248, -0.159]
GPT-4o-mini	0.895	0.743	-0.152	[-0.191, -0.114]

Source: run_review_response.json.

AgentDojo constructed benign + alternative fusion rules

system	AUC	diff vs classifier [CI]
input filter alone (AgentDojo)	0.968	
filter + GPT-4o-mini (AgentDojo)	0.908	
GLM-4.6 + filter, mean rule	0.922	+0.0065 [-0.014, 0.028] (best of all pairings)
GLM-4.6 + filter, product rule	0.916	+0.0002 [0.000, 0.001]

Source: run_review_response.json. Under mean or product fusion no pairing improves on the classifier alone on InjecAgent real benign (paper rounds the best diff, +0.0065, to +0.007).

InjecAgent loader contamination, same classifier (GLM-4.6, 510 records)

metric	pre-fix	post-fix	delta
AUC	0.518	0.915	+0.398
benign FPR @0.5	81.4%	0.0%	
TPR @7% FPR	1.1%	83.4%	+0.824

Source: run_contamination_prepost.json; cache glm_guard__injecagent_prefix.jsonl. Only the 323 benign rows differ between constructions.

Cheap text-derived signals (InjecAgent, 510)

signal	AUC	TPR@1%FPR
det-nonconformity	0.500	0.0%
gbdt-proba	0.500	0.0%
embedding-knn	0.501	2.7%
combined-max	0.504	0.0%

Source: run_signal_probe.json. Det nonconformity is identically zero - no signature fires on InjecAgent.

Adaptive benchmark: fused score under paraphrase attack

440 attacks / 240 benign, provider zai_glm46, bootstrap n=2000

system	AUC [95% CI]	TPR@7%FPR [CI]
text	0.778 [0.748, 0.808]	40.2% [0.3, 0.5]
provenance	0.627 [0.608, 0.649]	25.4% [0.2, 0.3]
pattern	0.651 [0.630, 0.673]	30.2% [0.3, 0.3]
text+provenance	0.782 [0.753, 0.812]	41.1% [0.4, 0.5]
text+pattern	0.791 [0.762, 0.822]	47.7% [0.4, 0.6]
provenance+pattern	0.699 [0.676, 0.723]	39.8% [0.4, 0.4]
text+provenance+pattern	0.795 [0.766, 0.825]	48.6% [0.4, 0.6]
lr@toolcall	0.795 [0.766, 0.825]	48.6% [0.4, 0.6]
lr@injecagent	0.781 [0.752, 0.812]	41.1% [0.4, 0.5]

Source: run_adaptive_fusion_eval_glm.json.

Key paired comparisons

comparison	AUC diff [95% CI]	McNemar p
text+provenance+pattern vs text+provenance	[0.007, 0.020]	0.00e+00
text+provenance+pattern vs text	[0.011, 0.026]	0.00e+00
text+provenance vs text	[0.001, 0.009]	1.25e-01

Detection by attack shape (fused 7% threshold)

tier	fused	without pattern channel
T.A	47.9%	39.6%
T.B	79.0%	66.0%
T.C	20.0%	20.0%

T.A tool-argument, T.B emitted-text, T.C data-flow. Per-family values follow on the next page.

Static vs adaptive (unpaired, indicative)

channel	static tool-call AUC	adaptive AUC	delta
text	0.832	0.778	-0.053
provenance	0.713	0.627	-0.085
pattern	0.656	0.651	-0.005
text+provenance	0.845	0.782	-0.062
text+pattern	0.861	0.791	-0.070
provenance+pattern	0.754	0.699	-0.055
text+provenance+pattern	0.874	0.795	-0.078

Per-family adaptive detection and taxonomy circularity

family	fused	no pat	family	fused	no pat
family	fused	no pattern	T.A.8	0.0%	0.0%
T.A.10	10.0%	10.0%	T.B.10	60.0%	5.0%
T.A.12	0.0%	0.0%	T.B.13	100.0%	100.0%
T.A.13	0.0%	0.0%	T.B.2	100.0%	100.0%
T.A.14	100.0%	100.0%	T.B.6	35.0%	25.0%
T.A.15	100.0%	0.0%	T.B.8	100.0%	100.0%
T.A.18	0.0%	0.0%	T.C.12	0.0%	0.0%
T.A.3	0.0%	0.0%	T.C.13	0.0%	0.0%
T.A.4	100.0%	100.0%	T.C.6	0.0%	0.0%
T.A.5	100.0%	100.0%	T.C.7	0.0%	0.0%
T.A.6	75.0%	75.0%	T.C.9	100.0%	100.0%
T.A.7	90.0%	90.0%			

Taxonomy circularity split (zero API)

draws tripping >= 1 signature	133 / 440 (30.2%)
draws fully evading signatures	307
fused detection on tripping draws	100.0%
fused detection on evading draws	26.4%
text detection on evading draws	25.1%
provenance flag rate on evading draws	13.7%
evading-draw detection by tier	T.A 26.5%, T.B 63.2%, T.C 0.0%
text scores <= 0.05 among attacks	263 / 440 (59.8%)
pattern channel cost	26.17 ms / trajectory, local
thresholds (pattern / fused @7% FPR)	0.6 / 0.1

Source: run_taxonomy_circularity.json, regenerated by run_taxonomy_circularity.py. The 133 tripping draws are exactly the pattern channel's TPR@7%FPR: detection is 100% on tripping draws and 0% on evading ones - the adaptive margin is a property of the draw distribution.

McNemar / Holm detail: subset pairs and adaptive

Holm-corrected subset-pair McNemar at matched 7% FPR: 16/21 toolcall, 18/21 injeagent, 2/3 agentdojo

pair	tc b	tc c	tc holm	InjA b	InjA c	InjA holm
text vs text+pattern	22	0	sig	0	0	-
text vs provenance+pattern	37	1	sig	15	175	sig
text vs text+provenance+pattern	37	0	sig	15	0	sig
provenance vs provenance+pattern	20	0	sig	0	0	-
provenance vs text+provenance+pattern	21	0	sig	175	0	sig
pattern vs text+pattern	33	0	sig	180	0	sig
pattern vs provenance+pattern	47	0	sig	20	0	sig
pattern vs text+provenance+pattern	48	0	sig	195	0	sig
text+provenance vs text+provenance+pattern	20	0	sig	0	0	-
text vs text+provenance	17	0	sig	15	0	sig
text+provenance vs provenance+pattern	20	1	sig	0	175	sig
text+pattern vs text+provenance+pattern	15	0	sig	15	0	sig
text vs provenance	17	1	sig	15	175	sig
text+pattern vs provenance+pattern	15	1	sig	15	175	sig
pattern vs text+provenance	48	20	sig	195	0	sig
provenance vs pattern	20	47	sig	0	20	sig
text vs pattern	22	33	-	0	180	sig
provenance vs text+provenance	1	0	-	175	0	sig
provenance vs text+pattern	21	15	-	175	15	sig
text+provenance vs text+pattern	20	15	-	0	15	sig
provenance+pattern vs text+provenance+pattern	1	0	-	175	0	sig

b = rows only the second system catches; c = only the first. Holm-corrected at 0.05 within each benchmark's 21-pair family (AgentDojo: 3 non-degenerate pairs, 2 significant). Source: run_ablation_ci.json.

Adaptive bench: McNemar at matched operating points

comparison	b	c	p
text+provenance+pattern vs text+provenance	0	33	0.0
text+provenance+pattern vs text	0	37	0.0
text+provenance vs text	0	4	0.125

b = smaller-only catches, c = larger-only; b = 0 means disagreements are one-directional. Zero-API recompute from the shipped adaptive caches. Source: run_adaptive_mcnemar.json.

Statistical treatment: CIs, McNemar, diversity

Full-grid bootstrap CIs (AUC), n_boot=2000, FPR budget 0.07

subset	InjecAgent	tool-call	AgentDojo
text	0.972 [0.949, 0.990]	0.832 [0.798, 0.867]	0.923 [0.896, 0.948]
provenance	0.503 [0.487, 0.521]	0.713 [0.682, 0.743]	n/a
pattern	0.500 [0.500, 0.500]	0.656 [0.627, 0.685]	0.152 [0.111, 0.195]
text+provenance	0.944 [0.916, 0.970]	0.845 [0.813, 0.877]	n/a
text+pattern	0.972 [0.949, 0.990]	0.861 [0.830, 0.891]	0.237 [0.172, 0.303]
provenance+pattern	0.503 [0.487, 0.521]	0.754 [0.725, 0.786]	n/a
text+provenance+patte rn	0.944 [0.916, 0.970]	0.874 [0.845, 0.901]	n/a

Source: run_ablation_ci.json. TPR CIs and Holm-corrected McNemar outcomes are in the same file; the AgentDojo pattern inversion is stable: frac(replicates < 0.5) = 1.0, CI [0.111, 0.195].

Channel-error diversity (Kuncheva)

bench	pair	Q	disagree	double fault
injecagent	pattern provenance	0.997	3.9%	35.1%
injecagent	pattern text	0.666	35.3%	2.2%
injecagent	provenance text	0.322	37.2%	1.6%
toolcall	pattern provenance	0.947	14.0%	24.6%
toolcall	pattern text	0.963	11.5%	27.7%
toolcall	provenance text	1.000	3.8%	28.7%
agentdojo	pattern text	1.000	36.8%	33.2%

Source: run_diversity_stats.json.

Statistical treatment (cont.): backbone, McNemar

Backbone-variant Spearman (text channel)

bench	pair	spearman
injecagent	claude-haiku-4-5 glm-4.6-guard	0.685
injecagent	claude-haiku-4-5 glm-5.2	0.687
injecagent	claude-haiku-4-5 gpt-4o-mini	0.583
injecagent	glm-4.6-guard glm-5.2	0.746
injecagent	glm-4.6-guard gpt-4o-mini	0.595
injecagent	glm-5.2 gpt-4o-mini	0.579
toolcall	claude-haiku-4-5 glm-4.6-guard	0.601
toolcall	claude-haiku-4-5 glm-5.2	0.670
toolcall	claude-haiku-4-5 gpt-4o-mini	0.655
toolcall	glm-4.6-guard glm-5.2	0.652
toolcall	glm-4.6-guard gpt-4o-mini	0.788
toolcall	glm-5.2 gpt-4o-mini	0.657
agentdojo	claude-haiku-4-5 glm-4.6-guard	0.403
agentdojo	claude-haiku-4-5 glm-5.2	0.588
agentdojo	claude-haiku-4-5 gpt-4o-mini	0.432
agentdojo	glm-4.6-guard glm-5.2	0.276
agentdojo	glm-4.6-guard gpt-4o-mini	0.250
agentdojo	glm-5.2 gpt-4o-mini	0.418

McNemar counts (system vs comparator)

comparison	b	c	p
toolcall_det_vs_lg4	193	50	0.0
agentdojo_system_vs_glm-5.2	140	3	0.0
agentdojo_system_vs_gpt-4o-mini	32	9	0.0004
agentdojo_system_vs_claude-haiku-4-5	165	2	0.0
agentdojo_system_vs_lg4	66	2	3.55e-16

Paired differences vs the text channel alone

comparison	AUCs	diff [CI]
learned head (00F), injecagent	0.9746 vs 0.9718	+0.0028 [-0.004, 0.013]
learned head (00F), toolcall	0.8695 vs 0.8316	+0.0379 [0.018, 0.059]
transfer tc->inj	0.9766 vs 0.9718	+0.0049 [-0.003, 0.018]
transfer inj->tc	0.8447 vs 0.8316	+0.0131 [0.005, 0.023]
text+prov max-fusion, toolcall	0.8448 vs 0.8316	+0.0132 [0.005, 0.023]

Source: run_paired_head_diffs.py (zero-API). The head matches text on InjecAgent (CI crosses zero) and adds where channels carry signal (toolcall); the taxonomy-independent subset's toolcall gain excludes zero.

Enforcement layer, layer robustness, AgentDojo FPR

Blast-radius evaluation (capability-tier gateway)

n attacks	90
n irreversible	78
n read sensitive	12
n benign	90
prevention irreversible default pct	100.0
prevention irreversible granted pct	100.0
prevention read sensitive pct	0.0
prevention under output bypass pct	100.0
false hitl default pct	100.0
false hitl granted pct	100.0

Layer-level adaptive bypass (440 paraphrase attacks)

input_filter_bypass	3/10
input_filter_bypass_pct	30.0
context_guard_bypass	5/11
context_guard_bypass_pct	45.5

AgentDojo layer FPR (120 constructed benign)

layer	TPR	benign FPR
input filter	30.7%	4.2%
context guard	86.1%	67.5%
tool gateway	9.6%	32.5%
det output filter	10.0%	76.7%
GLM-4.6 alone	20.7%	0.0%
FULL-SYSTEM	92.5%	67.5%

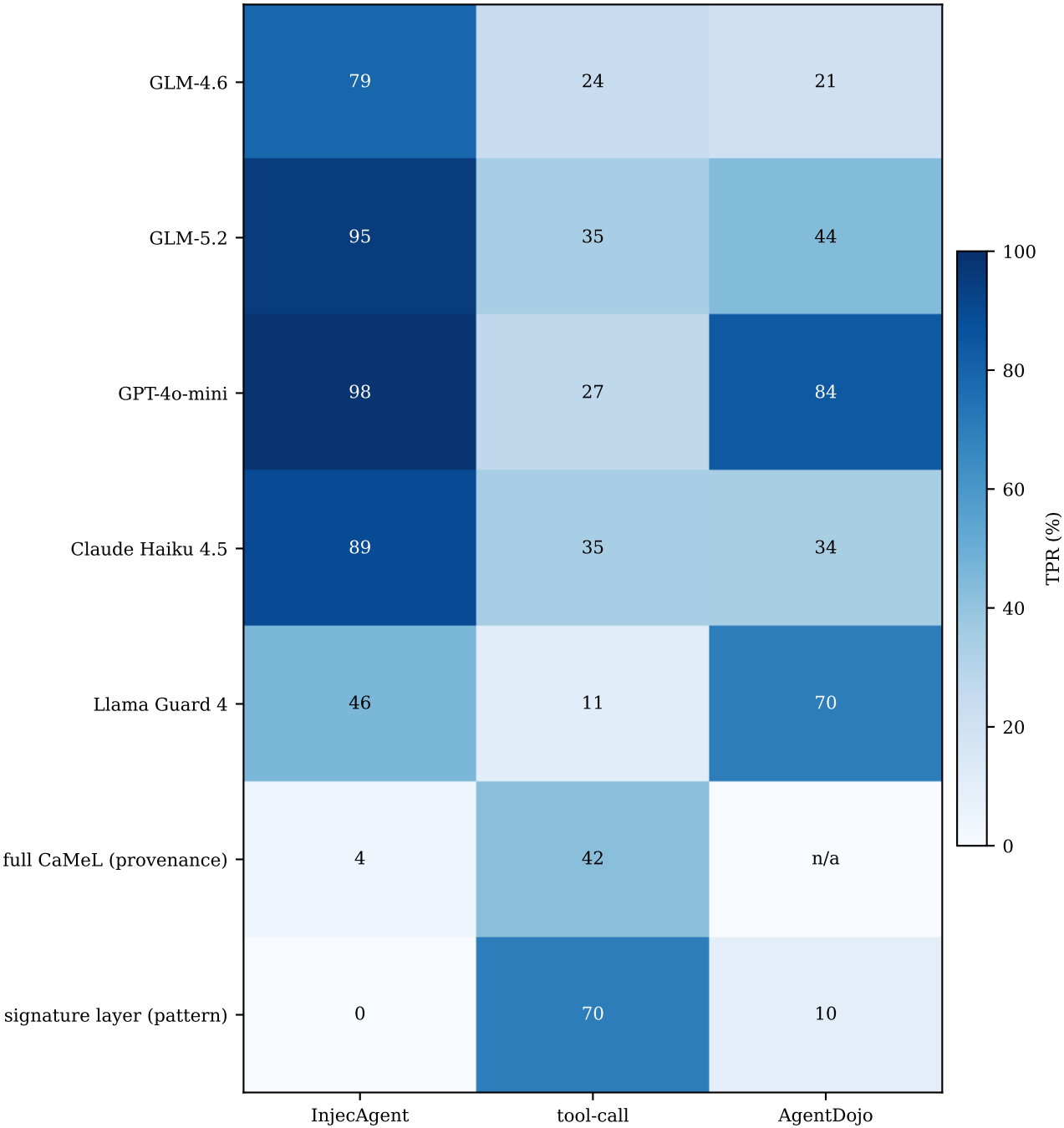
pii_benign_fraction = 0.675. Source: run_agentdojo_fpr.json.

Tool-call bench layer summary

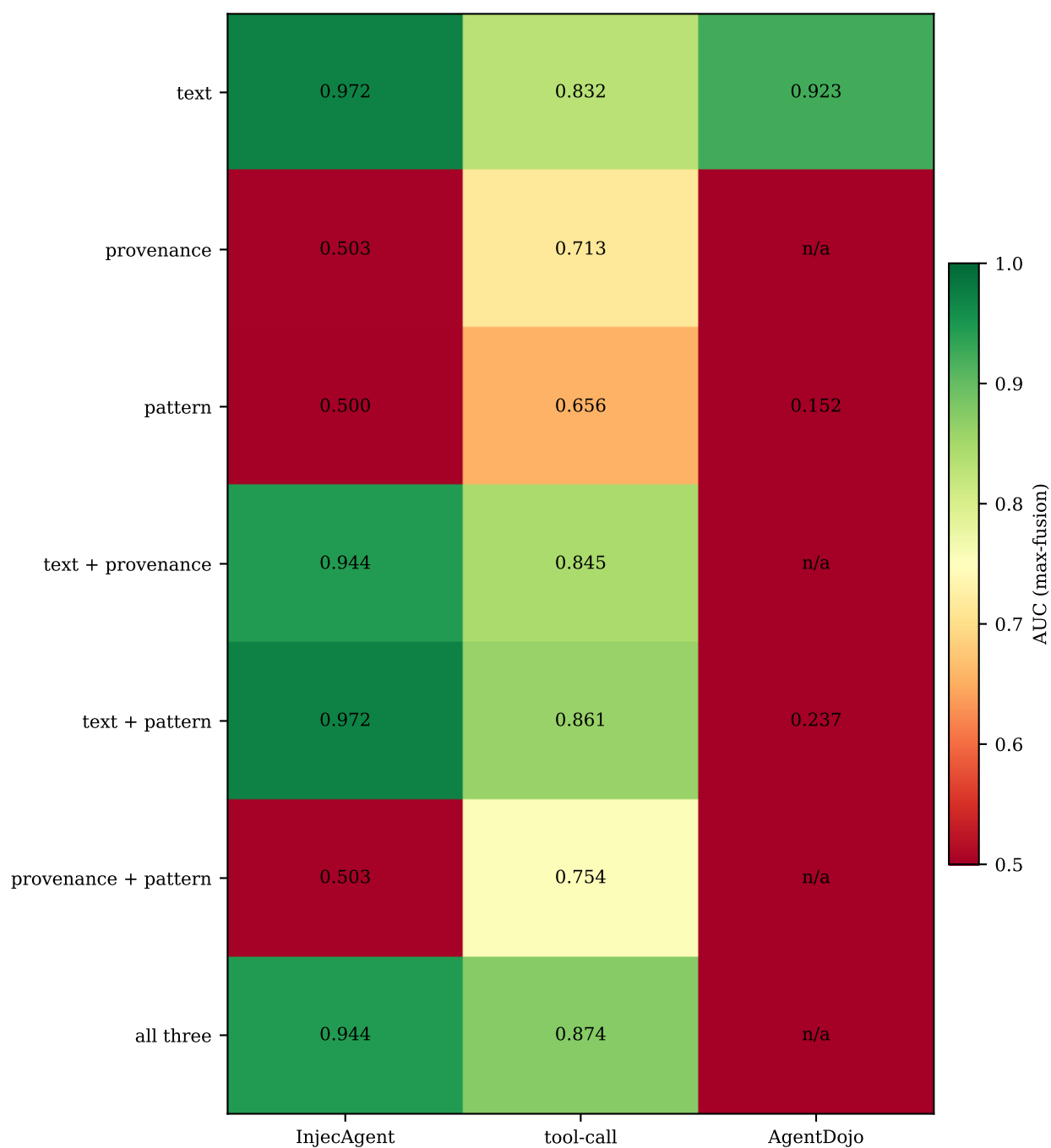
system	TPR [CI]	FPR [CI]
det-output-filter	70.0% [0.6, 0.8]	12.5% [0.1, 0.2]
GLM-alone (@0.5)	23.8% [0.2, 0.3]	0.0% [0.0, 0.0]
tool-gateway	64.6% [0.6, 0.7]	50.0% [0.4, 0.6]
input-filter	0.0% [0.0, 0.0]	0.0% [0.0, 0.0]
context-guard	0.0% [0.0, 0.0]	0.0% [0.0, 0.0]
FULL-SYSTEM (GLM)	74.2% [0.7, 0.8]	50.0% [0.4, 0.6]

GLM AUC on this bench: 0.642. Source: run_toolcall_bench.json.

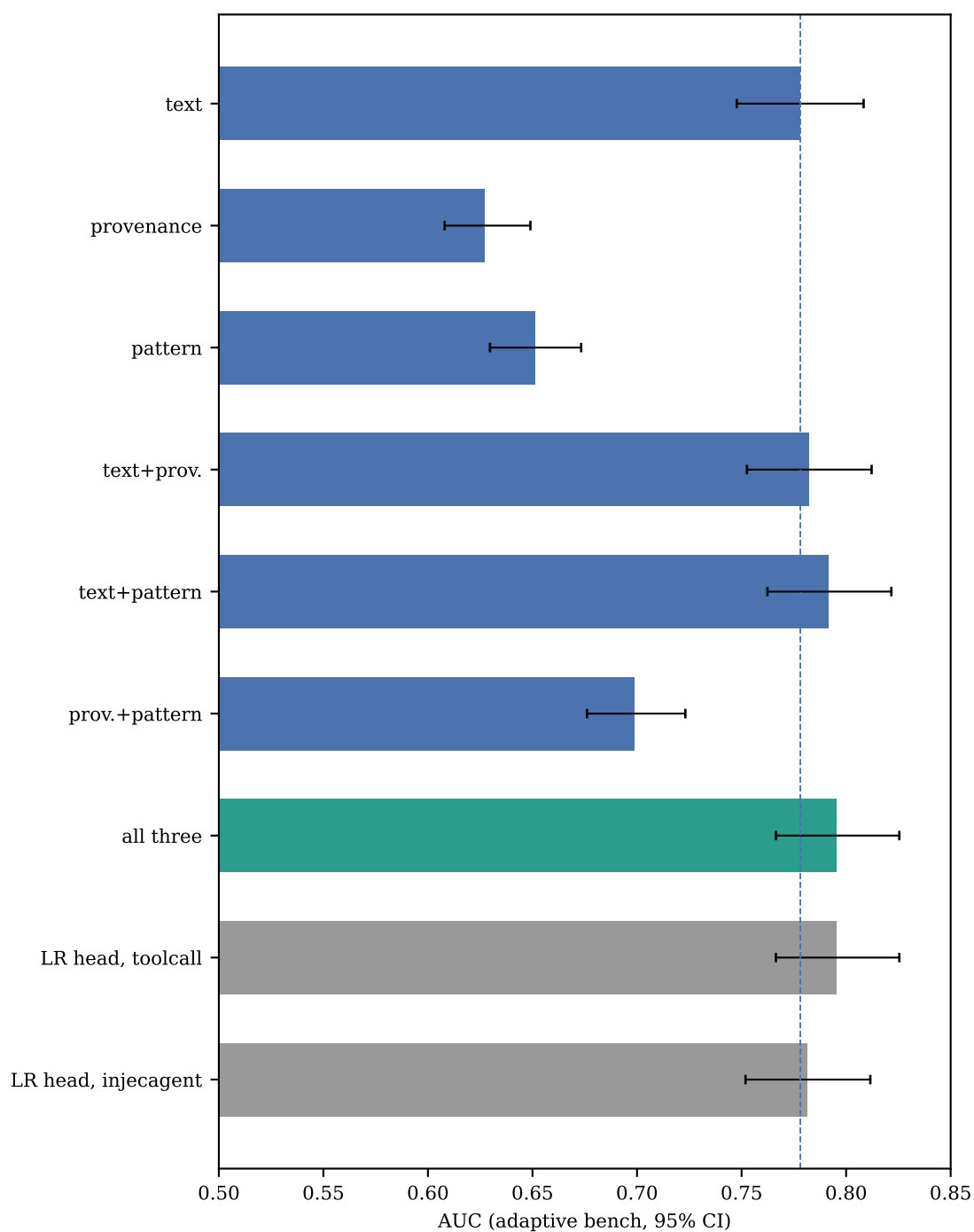
Figure: regime baseline heatmap



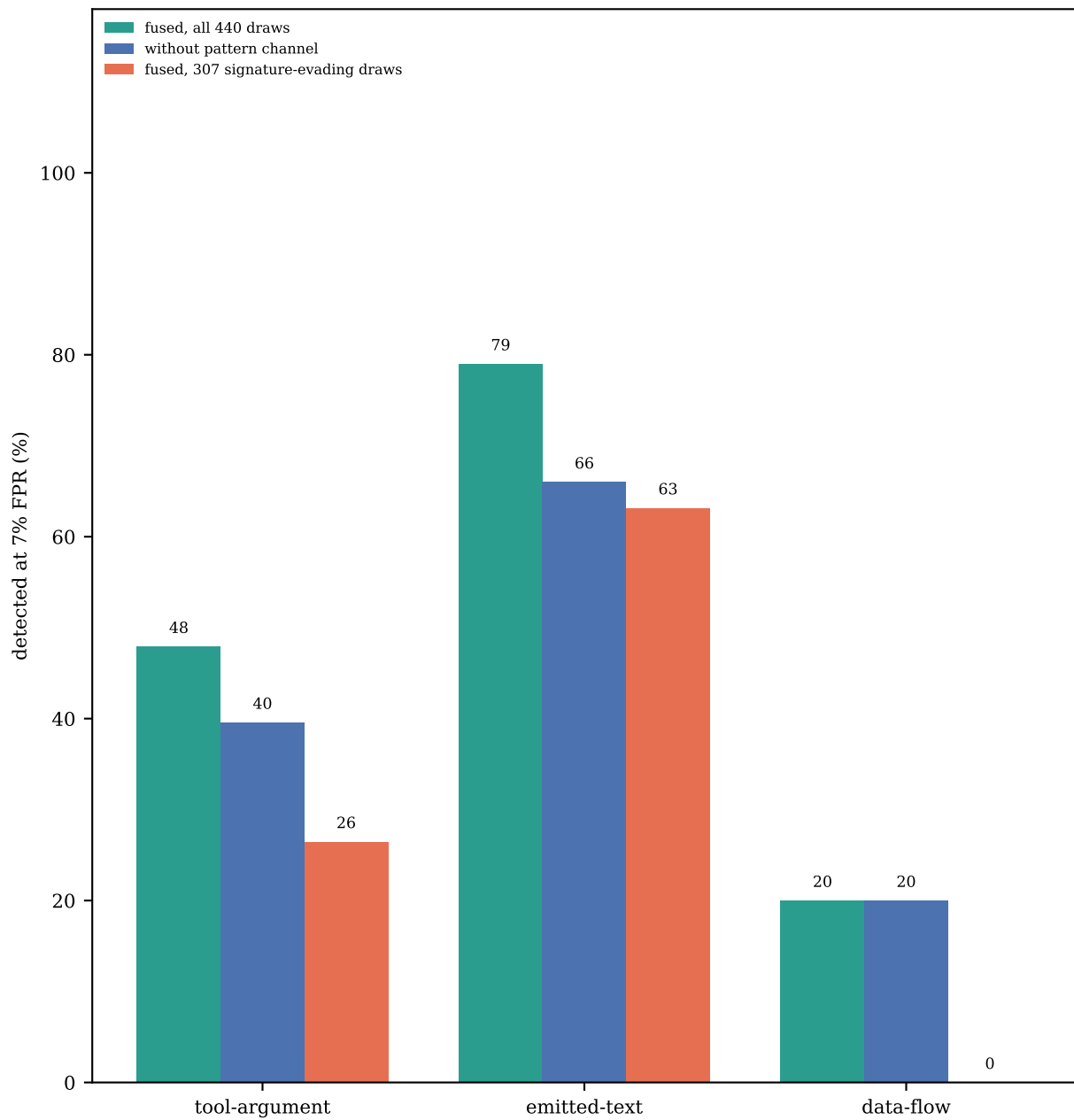
Classifiers are strong on text-shaped attacks and collapse on action-shaped ones; CaMeL is the mirror image; the signature layer is taxonomy-bound. Rendering identical to figs/make_figs.py (values from the regime table, page 2).

Figure: channel-subset ablation heatmap

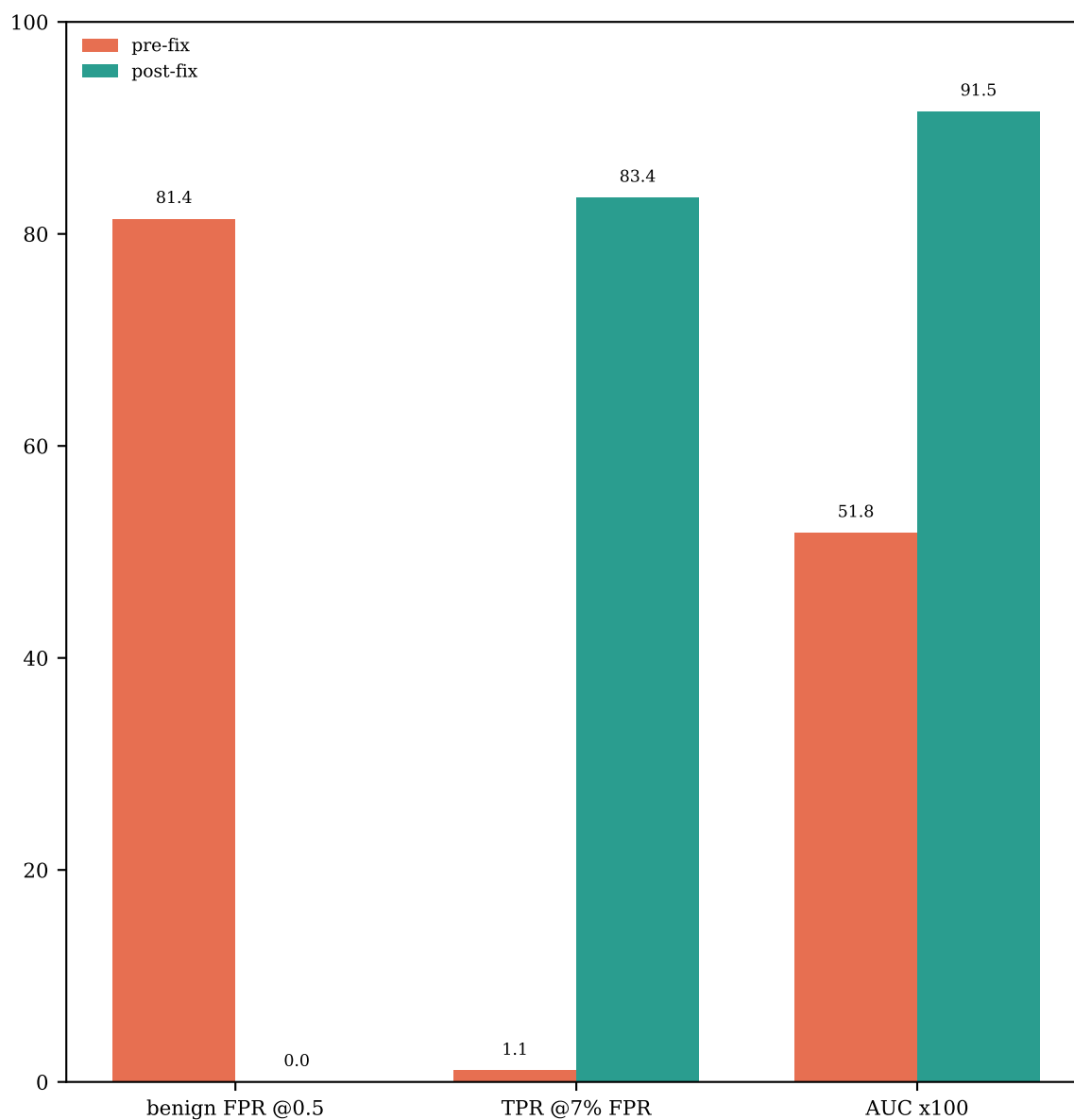
Monotone complementarity on the action-shaped bench; degradation below the best single channel where a channel is off-regime. Identical to figs/make_figs.py; AUC values equal run_ensemble_ablation.json (page 3).

Figure: fused margin under adaptation

Bars read live from run_adaptive_fusion_eval_glm.json (systems section, page 5). Dashed line: text channel alone.

Figure: detection by attack shape

First two bar groups read from `run_adaptive_fusion_eval_glm.json`; the evasion-conditional bars from `run_taxonomy_circularity.json`. Data-flow attacks are fully undetected once the signature layer is evaded.

Figure: contamination pre/post

Read from run_contamination_prepost.json: the loader bug put the text channel at chance (AUC 0.518) with 81.4% benign FPR; post-fix 0.916 and 0%.

Cost and latency per channel (Table VIII provenance)

Adaptive run: 440 attacks + 240 benign, one provider (GLM-4.6). Wall-clock and call counts are parsed from the shipped run log; prompt lengths and pattern timing are recomputed locally.

channel	calls/traj	wall-clock per traj	USD / 1k attacks
text (GLM-4.6)	1	2.07 s @ 8-way (16.58 s/call)	\$0.36
provenance (CaMeL)	2.36	17.71 s sequential (7.49 s/call)	\$1.63 (est)
pattern	0	27.1 ms local, single-thread	\$0
fused (max)	3.36	CaMeL-bound unless parallelized	\$1.99

Benign traffic (240 constructed benign)

CaMeL issued calls on only 18 of 240 benign trajectories (0.075 per trajectory, 0.51 s per trajectory amortized); text scored all 240 in 223 s at 8-way. Deployed benign-heavy traffic approaches text-plus-pattern cost.

Token and price provenance

Mean guard prompt 2098.3 characters over the 440 attacks (~525 input tokens at 4 chars/token; output is one score line). CaMeL per-call tokens estimated at 600 in / 150 out. September 2026 list prices; the same prompt on Claude Haiku 4.5 costs \$0.62 per 1,000. Pattern timing is machine-dependent (recomputed at render time).

Source: run_cost_table.py -> run_cost_table.json (parses run_adaptive_fusion_eval_glm.log; zero API calls).

Backbone deconfound (provenance re-scored on Gemma 4 31B)

The shipped tool-call channels share Claude Haiku 4.5 (text) and Haiku-backed CaMeL (provenance). This run re-scores ONLY the provenance channel on a different backbone, then recomputes every affected number from caches; all intervals are 95% paired bootstraps (n=2000). Source: `run_backbone_deconfound.json`.

Setup: n=480 tool-call trajectories, backbone google/gemma-4-31b-it for CaMeL P/Q (660 gateway calls, 659 answered); text stays on its shipped Haiku scores; pattern recomputed locally.

Manipulation check: provenance channel alone

provenance on	AUC	hard-label TPR	hard-label FPR
Gemma 4 31B (new)	0.708	4170.0%	0.0%
Haiku 4.5 (shipped)	0.713	-	-

Paired AUC difference (new - shipped): -0.0042 [-0.0107, 0.0000]

Fused triple and taxonomy-independent subset

system	AUC (new prov)	TPR@7% (new)	AUC (shipped prov)
text+prov+pattern	0.870	5000.0%	0.874
text+prov	0.841	4170.0%	0.845
text alone (unchanged)	0.832	-	-

Fused, new vs shipped provenance: -0.0040 [-0.0109, 0.0000] (TPR@7%: -0.0083 [-0.0214, 0.0000]).

Taxonomy-independent gain survives: text+prov (new) vs text alone +0.0092 [0.003, 0.017] AUC, +0.0625 [0.033, 0.094] TPR@7%.

Off-regime bench (InjecAgent, n=510): is the P2 degradation backbone-independent? 103 gateway calls, all answered (CaMeL policy short-circuits no-flow trajectories before any LLM call). Source: `run_backbone_deconfound_injecagent.json`.

system	shipped prov (Haiku)	new prov (Gemma)
provenance fires / trajectories	20 / 510	0 / 510
provenance alone AUC	0.503	0.500
fused triple AUC	0.944	0.972
fused TPR@7% FPR	9570.0%	9410.0%
LR head CV AUC	0.9776	0.9712 (text-only 0.9712)

Fused (Gemma prov) equals text alone exactly: paired AUC diff +0.0000 [0.0000, 0.0000] (constant column); fused new-vs-shipped +0.0277 [0.0050, 0.0501]. The degradation is the backbone's firing propensity: 20 hard fires import benign false positives; 0 fires import nothing.

Artifact inventory

file	bytes	top-level keys
all_mcnemar_counts.json	816	toolcall_det_vs_lg4, agentdojo_system_vs_glm-5.2, agentdojo_
run_ablation_ci.json	15,611	fpr_budget, n_boot, per_bench, agentdojo_pattern_sign_stabil
run_adaptive_fusion_eval_glm.json	5,206	n_attacks, n_benign_control, fpr_budget, n_boot, scoring_pro
run_adaptive_layer_robustness.json	3,013	input_filter_bypass, input_filter_bypass_pct, context_guard_
run_adaptive_mcnemar.json	1,002	config, systems, comparisons
run_agentdojo.json	1,647	bench, n_trajectories, systems, coverage
run_agentdojo_fpr.json	558	benign_n, attacks_n, pii_benign_fraction, layers
run_agentdojo_full.json	1,647	bench, n_trajectories, systems, coverage
run_agentdojo_matched_fpr.json	467	system_tpr, system_fpr, comparators
run_backbone_deconfound.json	2,033	config, provenance_single, fused_triple, text_plus_provenanc
run_backbone_deconfound_injecagent.json	2,018	config, provenance_single, fused_triple, text_plus_provenanc
run_blast_radius_eval.json	342	n_attacks, n_irreversible, n_read_sensitive, n_benign, preve
run_camel_full_eval.json	260	injecagent, agentdojo, toolcall
run_contamination_prepost.json	1,754	n, n_malicious, n_benign, note, pre
run_cost_table.json	1,435	config, measured_from_log, recomputed, cost_usd_per_1000_att
run_diversity_stats.json	2,692	fpr_budget, channel_diversity, backbone_correlation
run_ensemble_ablation.json	4,310	ablation, lr_fusion, transfer
run_fusion_rule_ablation.json	7,979	n_boot, single_channel_auc_ci, ablation_grid_max_ci, fusion_
run_injecagent_full.json	1,661	bench, n_trajectories, systems, coverage
run_lg4_bootstrap.json	4,419	baseline, model, base_url, note, reruns
run_lg4_comparison.json	2,418	injecagent, toolcall, agentdojo
run_lg4_mcnemar.json	184	toolcall_det_vs_lg4, agentdojo_system_vs_lg4
run_or_extra_comparison.json	4,851	glm-5.2, gpt-4o-mini, claude-haiku-4-5
run_paired_head_diffs.json	1,470	n_boot, comparisons
run_review_response.json	4,411	A_det_score_distribution, B_fusion_rule_reconstruction, B_co
run_signal_probe.json	1,290	bench, n_malicious, n_benign, signals, honest_fork_rule
run_taxonomy_circularity.json	17,182	n_attacks, pattern_threshold_7pct, fused_threshold_7pct, dra
run_toolcall_bench.json	1,397	bench, n, n_malicious, n_benign, glm_auc
run_toolcall_matched_fpr.json	969	bench, n, fprs, rows
run_tpr_at_1pct.json	2,450	config, ablation, paired_diffs, granularity

Per-trajectory caches (30 files, results/cache/):

camel_full_or_agentdojo.jsonl, camel_full_or_injecagent.jsonl, camel_full_or_toolcall.jsonl,
 camel_glm_adaptive.jsonl, camel_glm_adaptive_benign.jsonl, camel_gw_gemma_4_31b_it_injecagent.jsonl,
 camel_gw_gemma_4_31b_it_toolcall.jsonl, glm_guard_adaptive.jsonl, glm_guard_adaptive_benign.jsonl,
 glm_guard_agentdojo.jsonl, glm_guard_agentdojo_benign.jsonl, glm_guard_injecagent.jsonl,
 glm_guard_injecagent_prefix.jsonl, glm_guard_toolcall.jsonl, glm_guard_toolcall_benign.jsonl,
 lg4_agentdojo.jsonl, lg4_injecagent.jsonl, lg4_toolcall.jsonl, or_claude-haiku-4-5_agentdojo.jsonl,
 or_claude-haiku-4-5_agentdojo_benign.jsonl, or_claude-haiku-4-5_injecagent.jsonl, or_claude-
 haiku-4-5_toolcall.jsonl, or_glm-5.2_agentdojo.jsonl, or_glm-5.2_agentdojo_benign.jsonl,
 or_glm-5.2_injecagent.jsonl, or_glm-5.2_toolcall.jsonl, or_gpt-4o-mini_agentdojo.jsonl,
 or_gpt-4o-mini_agentdojo_benign.jsonl, or_gpt-4o-mini_injecagent.jsonl, or_gpt-4o-mini_toolcall.jsonl

Adaptive run log: results/run_adaptive_fusion_eval_glm.log (timing lines behind the cost table).